

IAGAPC-Enhanced: Kinematically Feasible Genetic Trajectory Optimization for Mobile-Sink Wireless Sensor Networks

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Abstract. Wireless sensor networks (WSNs) are equipped with limited resources, hence efficient use of energy is needed to increase lifespan of network. Inculcation of mobile sink to deal with energy issues improves the lifetime of the network. Mobile-sink trajectory planning is usually optimized for spatial coverage alone. It rarely asks whether the resulting path is the one a physical sink could execute efficiently. We propose IAGAPC-Enhanced (Improved Adaptive Genetic Algorithm with Path Compactness, Enhanced), a genetic algorithm that layers Catmull-Rom spline densification and an explicit kinematic-feasibility constraint onto adaptive operator control and a diversity-restart mechanism. Five baselines meet it on common ground: an adaptive-parameter GA, a tabu-search hybrid, an artificial-bee-colony hybrid, a distributed island-model GA, and a cluster-first K-means/TSP scheme. All six share one LEACH-style clustering and multi-hop relay substrate in ns-3.45, with ten random-initialization runs per scheme with confidence intervals called seeds. Across this evaluation, IAGAPC-Enhanced's trajectories turn substantially less deeper than every baseline's (22.99° vs. $74.81\text{--}109.89^\circ$, non-overlapping 95% confidence intervals). This is the one result our data separates cleanly. Packet delivery ratio, throughput, delay, and energy efficiency do not separate between schemes at this sample size. We report that directly rather than ranking the data does not support. Absolute delivery performance is low across all six schemes. We trace to PHY-layer contention among 200 nodes sharing a WiFi-modeled channel. We disclose that channel as a deviation from the IEEE 802.15.4 system model, not as equivalent to it.

Keywords: wireless sensor networks · mobile sink · genetic algorithm · trajectory optimization · cluster-head election · kinematic feasibility

1 Introduction

A wireless sensor network (WSN) with a single static base station forces nodes near that station to relay traffic for the rest of the field. Those nodes exhaust their

batteries first. This is the well-documented energy-hole problem. Introducing a mobile sink that physically visits the deployment area redistributes this load, but it converts a routing problem into a motion-planning problem. The sink’s trajectory now directly determines how much of the network it can service, and how often, before energy reserves run out.

Much of the literature settles the trajectory independently of the clustering and routing layer beneath it, or optimizes the path for spatial coverage alone. Either way, it never asks whether the resulting sequence of moves is something that a physical platform could execute. Geometric validity is not executability: sharp direction reversals burn travel time. Every second spent turning is a second the sink collects nothing, that cost shows up in the network’s energy budget and not the path length.

This paper targets that gap. We propose IAGAPC-Enhanced, a genetic algorithm for mobile-sink trajectory planning and combining adaptive crossover/mutation control, a diversity-restart operator, Catmull-Rom spline densification, and has explicit kinematic-feasibility constraint tying waypoint timing to the sink’s maximum velocity. We evaluate it against five baseline trajectory strategies representative of prior GA-based WSN work: an adaptive-parameter GA, a tabu-search hybrid, an artificial-bee-colony hybrid, a distributed island-model GA, and a cluster-first K-means/TSP scheme. All six run under one shared two-tier architecture: LEACH-style cluster-head election beneath the sink-trajectory layer, with greedy geographic multi-hop relay to the nearest live cluster head.

Our contributions are:

- A kinematically-constrained GA trajectory planner (IAGAPC-Enhanced) that produces markedly smoother sink paths than five comparably-structured baselines, evaluated under identical simulation conditions in all six cases.
- A ten-seed, six-scheme energy-efficiency comparison reported honestly as statistically inconclusive where the data supports no ranking, rather than a single favorable number presented without its overlapping confidence intervals.
- A fully instrumented ns-3 evaluation harness for six trajectory schemes sharing clustering/relay substrate, together with an explicit accounting of where the link layer, not the trajectory algorithm. This is the limiting factor for absolute delivery performance at this deployment scale.

Section 2 surveys related work by layer; Section 3 states the network and energy model; Section 4 develops IAGAPC-Enhanced; Section 5 details the simulation; results and conclusion follow in Sections 6–7.

2 Related Work

Genetic algorithms enter WSN design at several distinct layers, and we group prior work by layer rather than walk through.

A large share of this literature keeps the sink fixed and applies a GA to cluster-head selection or intra-network routing: encoding the whole source-to-sink route as a chromosome [9], weighing hop distance against residual energy in the fitness function [12], splitting sleep-scheduling and clustering into two sequential GA passes [1], or layering a trust score onto adaptive-GA routing for attack resilience [5]. B.D. Bala, H.M. Hassan, and S.M. Lawan [3] report a bacterial-conjugation operator beating a GA at the same clustering task in their setting. That is a reminder that genetic search is one viable metaheuristic here, not an automatic winner. G. Radhika and N. Radhika [15] replace GA-based election by pairing attribute-based cluster-head selection with a Q-learning routing algorithm embedded with action sequences. Adaptive Coati Optimization step tunes the algorithm’s action sequences to keep nodes from going unreachable. Their comparison is against three ablations of that same design: the optimized learner stripped of clustering, and the untuned learner run both with and without the result. These results report an 11.82% throughput gain and a 25.19% cut in energy consumption. None of this group moves the sink.

N.a. Mottaki, H. Motameni, and H. Mohamadi [11], A.K. Idrees and W.L. Al-Yaseen [7], S. Sun, Y. Chen, and L. Dong [19], A. Somauroo and V. Bassoo [18], and Varsha, M. Bala, M. Kumar, and N. Kumar [20] all use a GA to decide which deployed sensors stay active rather than to cluster or route. Their approaches include hybridizing tabu search into coverage-set scheduling, distributing election across virtual subregions, pairing a GA with reinforced whale-optimization for pure coverage, and replacing PEGASIS’s greedy chain with a GA-built one, tracing large lifetime gains to chain geometry specifically (a structural isolation that parallels what we do with trajectory smoothness). Varsha, M. Bala, M. Kumar, and N. Kumar route over a three-tier heterogeneous clustering protocol with a hybrid tabu-GA.

R. Hamidouche, Z. Aliouat, A.A.A. Ari, and A. Gueroui [4] propose work closest to our problem that optimizes the sink’s own path or stop points, steering a salp-swarm search toward the fullest cluster-head buffer over a fixed grey-wolf clustering. Another combines Voronoi deployment with a chaotic-Aquila path planner, beating six baselines on delay, lifetime, PDR, and throughput at once [17]. A third treats itinerary planning as a multi-criteria trade-off, centrally via IQCQP and distributedly via fuzzy TOPSIS; the gap between the two closes in dense deployments but opens to 14–26% (depending on the metric). Once the site grid thins out [8]. Others place Steiner-tree sink stops over bio-inspired clustering for a hybrid WSN/LTE path [10], and a GA-optimized cluster-head election is done explicitly for movable sinks, available to us only as an abstract [13]. Metaheuristics other than the GA, and learned policies, have also been aimed at the routing decision itself. G. Radhika, N. Radhika, and S. Paul [16] treat congestion as the thing to avoid scores. Each link with a traffic-density probability, builds an adaptive Petal Spider Ant Colony clustering over the result to pick stable next hops, and hand the surviving candidates to a Q-learning route selector. After which a proactive cluster-routing layer then throttles by a lookup on duty-cycle energy, spreading load to stretch lifetime. PAR, PSO, and

FAPRP are the yardsticks. Their scheme tops out at 99% on both routing and throughput performance with 100 nodes. The observed results are (84% and 83% respectively in the sparse 25-node case). It drops 15% of packets at 100 nodes and cuts end-to-end latency from 24 ms at 25 nodes to 8 ms at 100. A different tack appears in G. Radhika and N. Radhika [14]. Every sensor becomes an agent picking its own next hop by reward, and a spider-monkey optimization step nominates the candidates it chooses among, scoring them on residual energy and inter-node distance. Their NS-2 study is anchored on two baselines, MOSMO and Flock-CC. Against MOSMO they put numbers on exactly two metrics (latency and control overhead, both down 35%) and leave throughput, energy draw, network lifetime, and packet loss as plotted curves. No report exists pertaining to lifetime and delivery ratio as observed in the whole group measures. None reports a trajectory-smoothness or kinematic-feasibility figure we could set beside our own.

G.T. Al-Mamari, F. Bouabdallah, and A. Cherif [2] sit outside the GA-focused corpus entirely. They fix mobile-sink sojourn points by linear programming and cuckoo search over a grid topology, with an overhearing-aware energy model as their central contribution and no clustering or GA involved. We depart from it by placing trajectory optimization inside a GA framework, comparable operator for operator against the GA-family baselines above, and by evaluating with a packet-level simulator rather than an analytical grid model. Two further gaps shape our design. No paper surveyed here reports an offered-load sensitivity sweep, and we include one. None uses ns-3 either (NS2, MATLAB, and OMNeT++ dominate this literature); our choice forces explicit MAC/PHY contention modeling that an idealized channel would hide. A point Section 6 returns to. A controlled six-way comparison of distinct GA *family* structures under one shared network substrate, which is what we report here, does not appear elsewhere in this corpus.

3 System Model

N sensor nodes are deployed uniformly at random over a square field of side L , each tagged normal, advanced, or super. That tier fixes each node's initial energy budget, so the population is energy-heterogeneous from round zero. A single mobile sink follows a trajectory computed offline, before the run begins, as a fixed-length sequence of waypoints. That trajectory is exactly what the compared optimization procedures produce. Cluster heads are elected probabilistically every round, in the style of W.B. Heinzelman, A.P. Chandrakasan, and H. Balakrishnan [6]: a node i that has not served as cluster head in the previous $1/p$ rounds computes threshold

$$T(i) = \frac{p}{1 - p \cdot (r \bmod \frac{1}{p})} \cdot \frac{E_{\text{residual}}(i)}{E_{\text{initial}}(i)}, \quad (1)$$

(p = target election probability, r = round index) and becomes cluster head if a uniform draw falls below $T(i)$, with the residual-energy factor biasing election

away from already-depleted nodes. We chose this rule for its plainness. Richer alternatives score candidates on link quality, proximity, and energy jointly before handing routing to a learned policy [15]. Keeping election purely probabilistic instead guarantees that all six compared schemes inherit one identical, GA-independent clustering layer, so any difference between them traces back to the trajectory. Ordinary sensors reach their nearest live cluster head by greedy geographic relay: each hop picks whichever in-range neighbor is closest to the target, up to a fixed hop cap, and the simulator records a packet that runs out of forward progress rather than silently discarding it. Nothing in that hop choice estimates link stability, which is the quantity that stability-aware relay protocols use to steer around fragile links [16]. Distance to the target is our only criterion. Cluster heads receives and self-generated traffic to finite, drop-oldest queue, and flushes it to the sink. This is done only when the sink's trajectory brings it into its range. Figure 1 shows the resulting two-tier architecture.

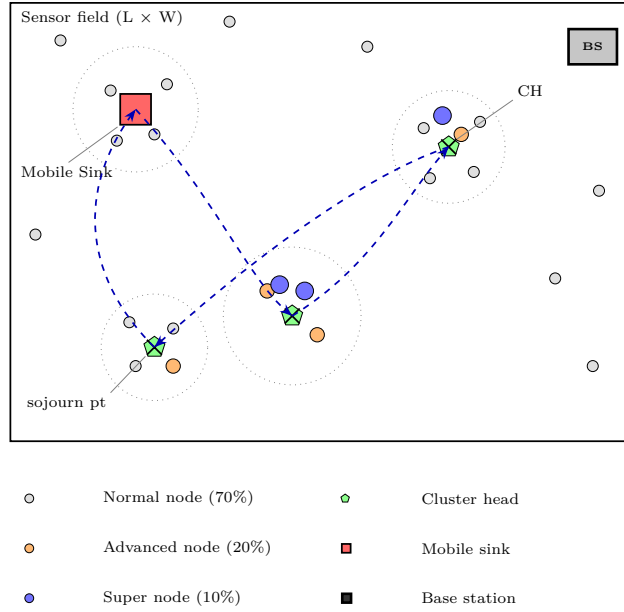


Fig. 1. Two-tier network architecture: ordinary sensor nodes (normal / advanced / super tiers) reach their nearest live cluster head via greedy multi-hop relay; cluster heads buffer aggregated traffic until the mobile sink's trajectory brings it within range.

Radio energy follows a current-draw mode, a fixed current per idle, receive, transmit, and sleep state, integrated over time spent in that state at a fixed supply voltage. A node is depleted once cumulative consumption reaches its tier budget. Radios default to sleep and wake only on a proximity triggers a

cluster head on sink proximity. An ordinary sensor on the existence of a live route to a cluster head that holds each sleep/wake window for a fixed duration (Table 1). These window lengths and per-state currents are this study’s own configuration choices, not values drawn from a cited source. Section 6 notes that this mechanism’s isolated contribution to the energy results could not be validated.

4 Proposed Methodology: IAGAPC-Enhanced

IAGAPC-Enhanced optimizes the sink’s trajectory as a fixed-length sequence of (x, y, t) waypoints via a GA with four features on top of a standard GA loop: adaptive operator rates, a diversity-triggered restart, spline-based densification, and kinematic-feasibility enforcement. Figure 2 summarizes the pipeline.

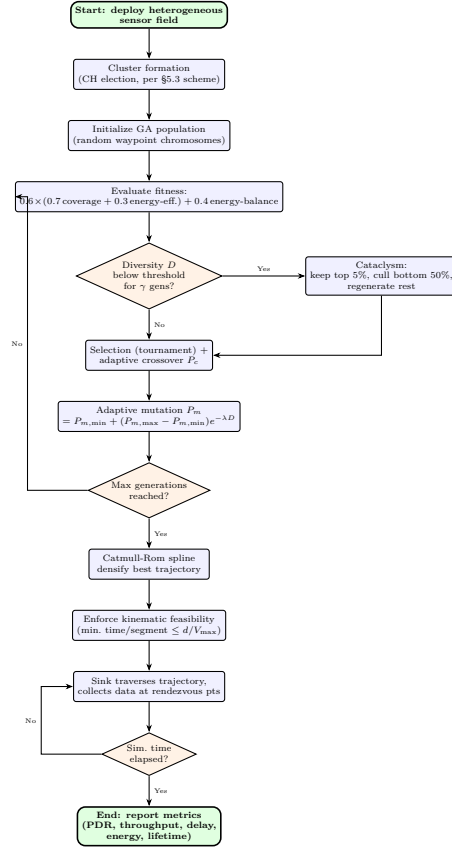


Fig. 2. Flow diagram of IAGAPC-Enhanced pipeline.

Fitness combines three terms: coverage (fraction of sensors within range of some waypoint), path length (shorter total distance), and compactness (shorter total duration against an a-priori target tour time, estimated from field geometry and sink speed via a Beardwood-Halton-Hammersley-style open-tour length). Compactness exists because neither of the other two terms depends on waypoint timing at all. Without it, nothing penalizes on visit times drifting toward the edge of their sampling range, and tour duration grows unbounded.

Each generation, crossover and mutation rates retune from the fitness spread (best minus average individual): a shrinking spread raises mutation pressure against premature convergence, a widening one relaxes it. When population diversity (mean waypoint-to-centroid distance) stays below threshold for several generations, a Cataclysm operator keeps only the incumbent best individual and reinitializes the rest, preventing the search from settling permanently into one basin.

Once the GA converges on control waypoints, Catmull-Rom spline interpolation inserts intermediate waypoints between each pair, producing a smoothly curving path instead of a piecewise-linear one. This is the mechanism directly responsible for Section 6’s trajectory-smoothness result. A final kinematic-feasibility pass increases any inter-waypoint time gap that would otherwise require exceeding the sink’s maximum velocity. The trajectory is physically executable, not merely geometrically valid. None of the five baselines apply this pass, reflecting the same sink-as-point-mass simplification common across the compared algorithm families.

5 Simulation Setup

Ns-3.45 harness drives all six schemes: clustering, delay, energy, and traffic behavior are byte-identical across runs (Table 1 gives the configuration). And only the trajectory-optimization procedure changes. One deviation from the system model is stated plainly rather than folded into the table: the implementation uses ns-3’s `WifiHelper` (IEEE 802.11) as the link layer, while Section 3 follows the IEEE 802.15.4 class in its range/rate assumptions. Deviation between implementation and specification, not presented as equivalent to it, whose measured effect on delivery is quantified in Section 6.

Table 1. Simulation parameters.

Parameter	Value
Simulator	ns-3.45
Deployment area	1000 m $\times$ 1000 m
Node count	Harness supports 100, 200, 300, 500; Section 6 results are $n=200$ only
Node distribution	Uniform random
Energy tiers	0.5 J / 1.0 J / 1.5 J at 70% / 20% / 10% of nodes
Link layer	IEEE 802.11b (WifiHelper), see deviation note above
Link data rate (PHY)	1 Mbps
Simulated time per run	$3\times$ a-priori tour-time target
Sensing-generation rate	1 packet / node / 10 s, fixed
Application payload	512 B per packet
Sensor-to-cluster-head range	50 m
Cluster-head-to-sink range	100 m
Cluster-head election probability p	0.05
Relay hop cap	3
Sink velocity	10 m/s
Duty-cycle check interval	1.0 s
Duty-cycle wake window	0.3 s on proximity trigger
GA population size	50
GA generations	50
GA waypoints	8
Seeds per configuration	10

6 Results and Discussion

All results below come from a single frozen configuration ($n=200$ nodes, 10 independent seeds per scheme, six schemes, base parameters from Table 1) runs after every fix described in Section 5 was in place. No result in this section predates that configuration. Reported intervals mean $\pm$ 95% confidence interval across the 10 seeds.

6.1 Trajectory Smoothness

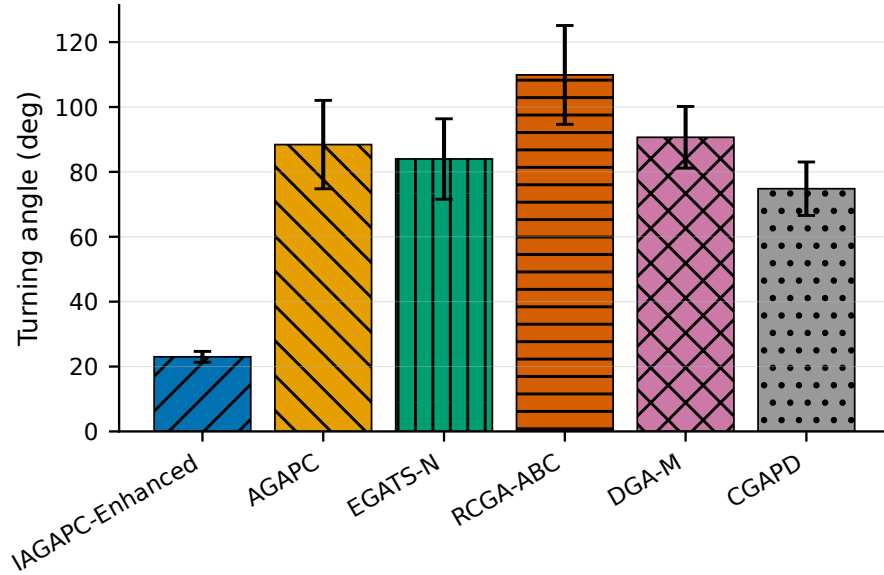
Table 2 reports the average absolute turning angle between consecutive trajectory segments. The geometric smoothness metric directly targeted by IAGAPC-Enhanced’s spline densification and kinematic-feasibility pass. This is also the one metric in our evaluation that is purely a function of the waypoint geometry. Each scheme outputs does not depend on the network, MAC, or PHY layer at all, which is why it is the result we treat as primary.

IAGAPC-Enhanced’s confidence interval does not overlap any baseline’s that narrow gap is to CGAPD, whose mean sits more than 45° away, with no shared range. The mechanism is directly attributable to the spline-densification and

Table 2. Average turning angle by scheme (degrees, mean $\pm$ 95% CI, $n=200$, 10 seeds).

Scheme	Turning angle ($^{\circ}$)
IAGAPC-Enhanced	22.99 ± 1.67
CGAPD	74.81 ± 8.23
EGATS-N	83.97 ± 12.39
AGAPC	88.41 ± 13.62
DGA-M	90.66 ± 9.50
RCGA-ABC	109.89 ± 15.23

kinematic-feasibility steps described in Section 4. No other scheme applies either, and no other scheme comes close to this result. This is the cleanest, most reproducible finding in our evaluation.

**Fig. 3.** Turning angle by scheme (mean $\pm$ 95% CI, $n=200$, 10 seeds).

One bar stands well below the rest. IAGAPC-Enhanced's whisker stops short of every baseline whisker, which happens on no other chart in this section, and the five baselines occupy a band from 74.8° to 109.9° while IAGAPC-Enhanced holds near 23° .

Table 3. Energy efficiency by scheme (packets/J, mean $\pm$ 95% CI, $n=200$, 10 seeds).

Scheme	Packets / Joule
EGATS-N	2.52 ± 1.10
AGAPC	3.13 ± 2.33
IAGAPC-Enhanced	4.33 ± 1.98
RCGA-ABC	5.16 ± 6.41
CGAPD	8.29 ± 6.38
DGA-M	8.39 ± 7.84

6.2 Energy Efficiency

Unlike smoothness, energy efficiency depends on the whole network stack, routing, contention, and duty-cycling all feed into packets delivered per joule. Table 3 gives this ratio over the same configuration.

Confidence intervals overlap substantially across all six schemes. DGA-M's alone spans 0.55 to 16.22 packets/J. At this sample size the six schemes are statistically indistinguishable on this metric. We report that rather than selecting the numerically highest mean as a headline claim.

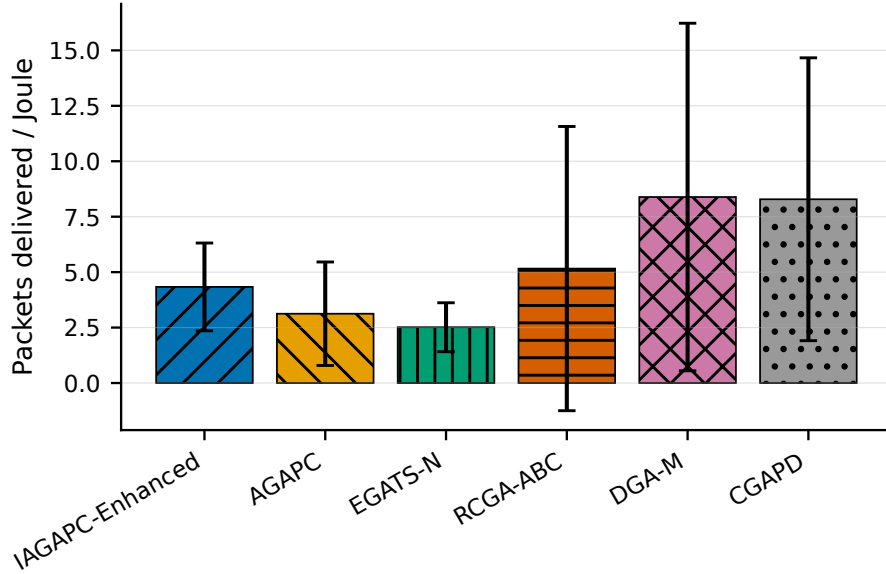
**Fig. 4.** Energy efficiency by scheme (packets/J, mean $\pm$ 95% CI, $n=200$, 10 seeds).

Table 4. Delivery ratio, throughput, and delay by scheme (mean $\pm$ 95% CI, $n=200$, 10 seeds).

Scheme	PDR (%)	Throughput (kbps)	Delay (s)
IAGAPC-Enhanced	0.071 ± 0.031	0.058 ± 0.026	11.86 ± 15.21
AGAPC	0.051 ± 0.038	0.042 ± 0.031	11.50 ± 14.90
EGATS-N	0.041 ± 0.018	0.034 ± 0.015	6.16 ± 5.43
RCGA-ABC	0.084 ± 0.105	0.069 ± 0.086	8.87 ± 16.98
DGA-M	0.136 ± 0.127	0.112 ± 0.105	19.77 ± 23.77
CGAPD	0.135 ± 0.104	0.111 ± 0.085	41.62 ± 35.41

6.3 Delivery Ratio, Throughput, and Delay

PDR ranged 0.041% (EGATS-N) to 0.136% (DGA-M) with CIs overlapping between every pair. Throughput and delay show the same pattern. Neither metric separates the six schemes at this sample size, which we report directly rather than an unsupported ranking. Section 6.5 explains why absolute delivery is low across all six at once. The more informative reading of this result than any relative ordering. We tabulate the underlying per-scheme means and 95% CIs in Table 4.

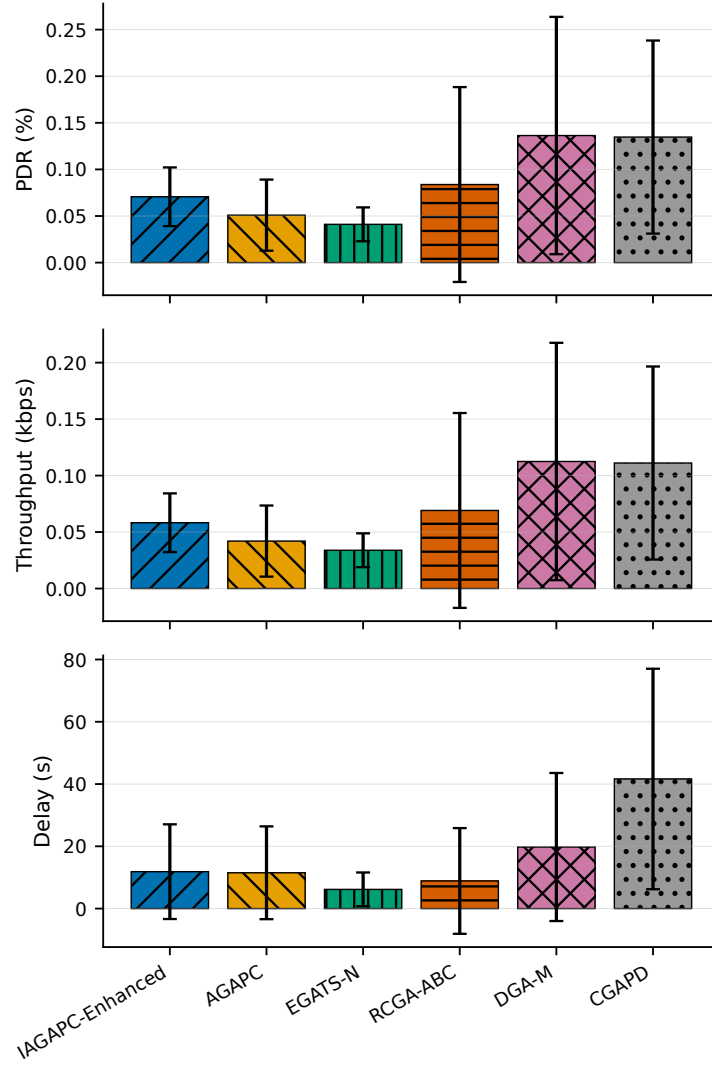


Fig. 5. PDR, throughput, and delay by scheme (mean $\pm$ 95% CI, $n=200$, 10 seeds).

The plotted intervals repeat what Table 4 lists numerically: on each of the three panels, all six schemes overlap. CGAPD's delay interval is the extreme case at 41.62 ± 35.41 s, wide enough to swallow most of the other point estimates. What changes from panel to panel is the absolute scale, not the separation between schemes.

Table 5. Aggregate loss breakdown, 60 runs (6 schemes $\times$ 10 seeds), as % of 610,094 relay-hop sends.

Outcome	Count	%
Received at sink	630	0.10%
Greedy-forwarding dead end	129,053	21.15%
Leaf buffer overflow (drop-oldest)	30,100	4.93%
Stranded, cluster head never sink-visited	4,667	0.76%
Stranded, cluster head visited but not yet flushed	195	0.03%

6.4 Network Lifetime: Evaluation Gap

The energy recalibration needed (Section 5) for nodes to realistically survive showed full sink tour that eliminated node mortality within the simulated horizon, across all 60 runs, FND/HND/LND are all unreachd. We report no values for them, since none exist at this horizon. An evaluation gap requiring a longer simulated duration to close, not a finding about scheme behavior.

6.5 Loss Breakdown, PHY-Layer Contention, and Limitations

Absolute delivery performance is low across every scheme in this evaluation (Section 6.3), and we traced why rather than leaving it unexplained. Table 5 decomposes 610,094 relay-hop-level transmissions, aggregated across all 60 runs of this configuration, by outcome.

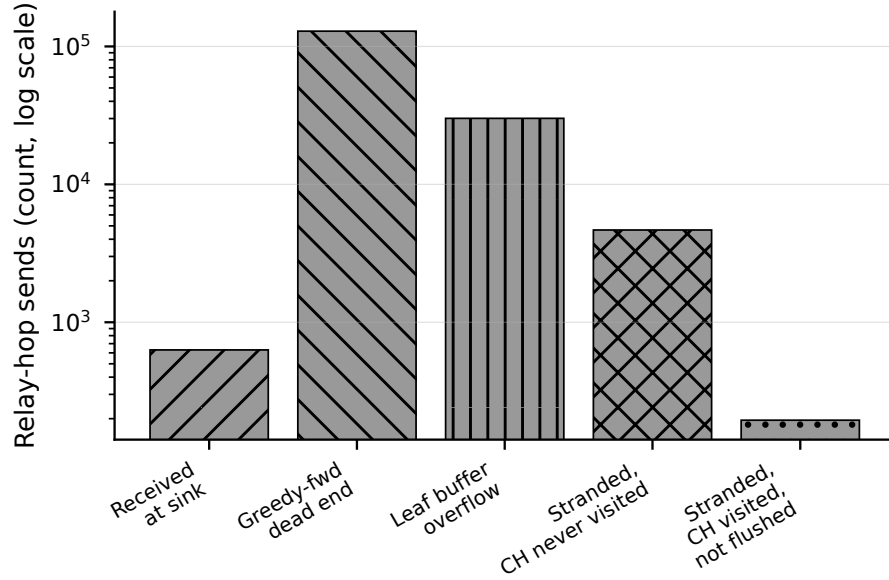


Fig. 6. Loss breakdown by outcome, 60 runs aggregate (log scale; outcome counts span 195 to 129,053).

The log axis is not cosmetic. Greedy-forwarding dead ends number 129,053 against 195 for the smallest stranded category, nearly three orders of magnitude apart. A linear axis would compress every bar except the dead-end and leaf-overflow ones onto the baseline. Those two outcomes take most of the losses we can attribute. The 630 packets that reached the sink form the second-shortest bar.

These five outcomes cover roughly 27% of relay-hop sends; the rest is invisible to any application-layer counter. NS-3’s own PHY-layer drop trace explains why, it fires 7,754,706 times across the same 60 runs, over twelve times the total sends. It counts every reception failure at every radio in range, not only packets addressed to it. PHY-layer contention among 200 nodes on one channel under greedy multi-hop relay is the dominant loss mechanism, and it traces to a disclosed system-model deviation: **WifiHelper** (IEEE 802.11b, lowest rate), not IEEE 802.15.4, not presented as equivalent. RTS/CTS plus the lowest data rate cut PHY reception failures 32% without resolving the underlying contention. It is consistent with a property of the 802.11 contention model at this density, not a tunable parameter. A narrowband CSMA-CA 802.15.4 PHY would not necessarily reproduce this severity, and we make no claim it would. Absolute PDR, throughput, and delay here characterize this WiFi-modeled layer under 200-node contention, not a real 802.15.4 deployment.

Two further disclosed limitations. Our current-times-airtime energy accounting agrees with the classic rate-independent first-order model only in a narrow

data-rate band. At our low sensing rate, per-packet energy cost runs roughly an order of magnitude above that formula’s prediction. The proposed energy figures are not interchangeable with classically-computed lifetime results. Separately, duty cycling’s isolated contribution to Table 3 could not be validated. It disables the only code path that transmits anything, giving a degenerate zero-packet comparison, not a meaningful ablation. Neither limitation confounds the six-scheme comparison itself. Every run applied identical configuration across all six, so both bound only the interpretation of absolute magnitudes.

Across Tables 2, 3, and 4: turning angle separates cleanly (IAGAPC-Enhanced vs. all five baselines, non-overlapping CIs); PDR, throughput, delay, and energy efficiency that do not (overlap 95% CIs). Network lifetime is unevaluated at this horizon (Section 6.4); Table 5 (Section 6.5) traces why absolute delivery is low across all six.

7 Conclusion and Future Work

We proposed IAGAPC-Enhanced, a GA for mobile-sink trajectory planning adding spline densification and kinematic-feasibility enforcement to adaptive operator control and a diversity-restart mechanism, and evaluated it against five baselines under one shared clustering, relay, and energy substrate in ns-3.45.

One finding is clean and statistically separated across ten seeds, IAGAPC-Enhanced’s trajectories turn substantially less sharply than any baseline’s (22.99° vs. $74.81\text{--}109.89^\circ$, non-overlapping CIs), directly attributable to its spline-densification and kinematic-feasibility mechanisms. PDR, throughput, delay, and energy efficiency do not separate the six schemes at this sample size, and we report that rather than an unsupported ranking. Absolute delivery is low across all six, traced (Section 6.5) to PHY-layer contention among 200 nodes on one WiFi-modeled channel. This results in a disclosed deviation from the IEEE 802.15.4 system model, not presented as equivalent to it. It is identical across all six schemes, so it explains none of the (absent) between-scheme difference.

Three limitations motivate future work: the WiFi-modeled link layer (`LrWpanHelper` would place contention under the actual target radio class), network lifetime unevaluated at this horizon, since the energy recalibration needed for tour survival also eliminated mortality within it. And a single sink’s hard coverage ceiling over a 200-node, $1000\text{ m}\times 1000\text{ m}$ deployment, which a multi-sink extension would relax, letting trajectory-quality differences surface in delivery metrics where one sink’s ceiling currently masks them.

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